DHOLE PATIL EDUCATION SOCIETY'S

DHOLE PATIL COLLEGE OF ENGINEERING

Department of First year Engineering

Course Code: ESC-102-ELE | **Subject: Basic Electrical Engineering** | **Academic Year – 2025-26**

Assignment - II | **Semester – I & II**

Unit –II Electromagnetism		**CO**	**BT Level**	**Marks**
S.NO:	**QUESTIONS**			
Que.1	Define a) Reluctance & its unit b) Permeance and its unit. c) Magnetic flux d) Magnetic flux e) Flux density f) Magnetic field intensity.	CO2	01	06
Que.2	Define Permeability. And derive expression for $\mu = \mu_r * \mu_0$	CO2	02	06
Que.3	State Faraday's law of Electromagnetic Induction.	CO2	01	06
Que.4	Explain with neat diagram Self Induced EMF and derive the expression for self inductance	CO2	02	06
Que.5	Compare Electric circuit and Magnetic circuit.	CO2	04	06
Que.6	A coil of 500 turns and resistance 20 Ohm is wound uniformly on an iron ring of mean circumference 50 cm and cross-sectional area $4cm^2$. It is connected to a 24 V d. c. supply. Under these conditions, the relative permeability of iron is 800. Calculate the values of (i) Magnetomotive force of the coil (ii) Magnetising force (iii) Total flux in iron (iv) Reluctance of the ring	CO2	03	06
Que.7	A coil is wound with 600 turns on a steel ring of relative permeability of 1600. It has a mean circumference of 30 cm and cross sectional area of 9 cm^2. If the coil has resistance of 20 ohm and it is connected to 20 V DC supply. Calculate: (a) Current in the coil (b) Reluctance (b) Field Strength (c) MMF (d) Total Flux	CO2	03	06
Que.8	Obtain the expression for the mutually induced emf in a coil and derive the formula for mutual inductance	CO2	02	06
Que.9	Derive expression for coefficient of coupling.	CO2	03	06
Que.10	A magnetizing force of 8000 A/m is applied to a circular magnetic circuit of mean diameter 30 cm by passing a current through a coil wound on the circuit. If the coil is uniformly wound around the circuit and has 750 turns, find the current in the coil.	CO2	05	06